

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Spring 2025

Course Title: Data Structures and Algorithms I

Course Code: CSE0613211

Marks: 50

Time: 3(three) hours

(Answer all questions)

Q. No.

Marks

[05]

1. a) For the following array, devise an algorithm to build a max-heap and show the final result.

45	28	52	25	60	70
1	2	3	4	5	6

Fig 1 - An Array representing a tree.

Here, for a node in the i -th position, the left child refers to the $(2*i)$ -th position and right child refers to the $(2*i + 1)$ -th position.

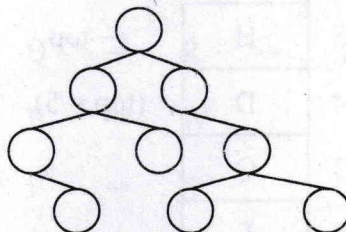


Fig 2 - A binary Tree

Construct an algorithm and show the results of traversing the above-mentioned binary tree in a post-order traversal manner.

2. a) Explain the following statement with proper reasons and examples- "Every tree is a graph, but not every graph is a tree."

[02]

3. a) Consider the following Double Ended Queue and apply the mentioned operations. Demonstrate the intermediate results after each step with proper explanation and visualization.

[03]

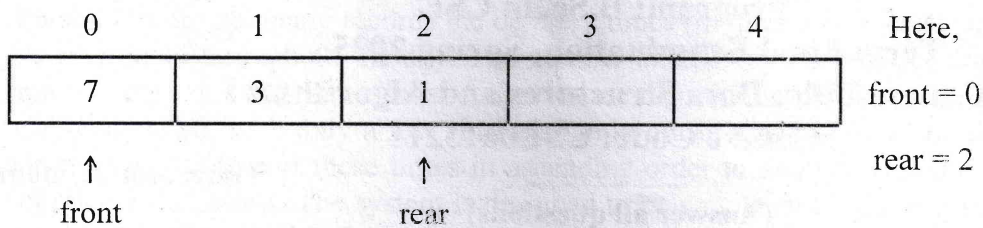


Fig 3 - A Double Ended Queue

- Insertion of 58 at the front end.
- Deletion at the front end.
- Deletion at the front end.
- Deletion at the rear end.
- Deletion at the rear end.
- Deletion at the rear end.

b) Write down an algorithm to pop an element from the following stack and apply the algorithm three times on the following stack. [04]

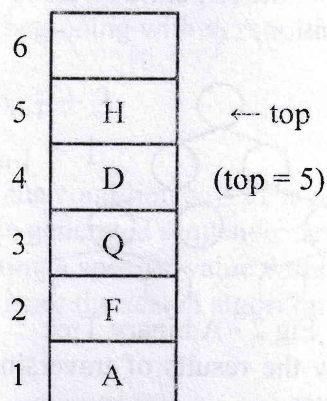


Fig 4 - A stack with a size of 6.

c) A factory machine is responsible for processing incoming tasks submitted by various departments. Each task is labeled with an urgency level represented by an integer, where a lower number means higher urgency. The machine must always process the most urgent task available. New tasks can arrive at any time, and the machine should be able to efficiently determine which task to execute next. [03]

Analyze the scenario and name a data structure that is most suitable for managing the tasks in this scenario, with an explanation.

	A	B	C	D	E	F
A	0	1	0	0	0	0
B	0	0	1	0	0	0
C	1	0	0	1	0	0
D	0	0	0	0	1	1
E	0	0	0	0	0	0
F	0	0	0	0	1	0

Fig 5. A matrix representing a Graph

- Create a graph from the above-mentioned matrix.
- Explain whether the given graph is directed or undirected based on the adjacency matrix.

b) Develop a graph from the following representation with Linked Lists and Devise an algorithm that will perform the best in traversing the graphs starting with Node P. [06]

Vertices List of Adjacency Vertices

P → Q → R → S → T
 Q → P → W
 R → P → X
 S → P → Y
 T → P → Z
 W → Q → A
 X → R → A
 Y → S → A
 Z → T → A
 A → W → X → Y → Z

Fig 6 - Linked representation of Adjacency Lists of a Graph.

4. a) A small courier company records the delivery times (in minutes) of each package as they are completed throughout the day. The number of delivery records per day is relatively small (usually < 100). Since most deliveries are recorded in chronological order, the list is almost sorted, with only a few entries slightly out of place. At the end of the day, the company needs to sort these times in ascending order to analyze which deliveries were completed the fastest. The system is designed to be simple and lightweight, and memory usage needs to be minimal. [07]

- Based on the scenario described above, which sorting algorithm would be most suitable for sorting the delivery times? Justify your answer.
- Using the delivery time list [22, 25, 20, 27, 30, 19, 28], demonstrate step by step how your chosen algorithm would sort these values.
- If the delivery system is later upgraded to handle over 10,000 delivery records per day, would the same sorting algorithm still be a suitable choice? Interpret your reasoning clearly.

- b) Explain the folding method of hashing with an example. [03]

5. a) Illustrate a numerical sequence beginning with $a_1 = 6$, where for all $n \geq 2$: [05]

- If n is even, then $a_n = a_{n-1} + 3$
 - If n is odd, then $a_n = a_{n-1} + 1$
- i. Construct the sequence until the condition $a_n \geq 21$ is satisfied.
 - ii. Based on the properties of the generated sequence, determine the most appropriate searching algorithm for locating a specific value within it.
 - iii. Justify your selection by devising the search algorithm step by step.

- b) You're implementing a hash table for utilizing memory shortage. [05]

- Initial table size (m) = 17
- Hash functions: $h_0(k) = k \bmod m$

Explain whether these functions are sufficient enough to store the following values in the memory. If not, suggest a solution.

The values: 18, 19, 28, 35, 52, 69.